QISKIT FALL FEST 2026

ANTIGRAVITY MASTER INSTRUCTIONS

A beginner-to-submission execution plan for Phase 1

Purpose: Give Antigravity one structured task brief for inspecting your existing work, using the official competition materials, implementing the selected problem, running Processor A and B, producing evidence, and preparing a separate GitHub submission repository.

Project context: Existing project: FloquetSim Explorer. Previously discussed problem: S3 — Noise in Trotter Simulation. Continue with S3 only if that is the problem you officially selected/registered.

Important truth: Antigravity can write and run code, but it cannot legitimately invent missing official processor definitions, fabricate measurements, or guarantee a competition submission without any verification. This guide instructs it to verify outputs and clearly report blockers. You should at least review the final checklist before submitting.

Official materials referenced in this guide

Phase 1 Participant Instructions supplied in this conversation; and the organizer Starter Guide folder: https://drive.google.com/drive/folders/1caxlXXeqjOkLmXGlimy_Etw0uPHSwyga

1. Competition requirements Antigravity must obey

The official Phase 1 instructions say participants must select exactly one problem statement, implement it, run the same controlled experiment on Processor A and Processor B using the exact organizer-provided definitions, compare the results, and submit a reproducible GitHub repository. The required package includes code/notebook, processor definitions or required files, results, README, environment specification, graphs, and scientific analysis. The instructions state that Processor C and custom Processor D belong to Phase 2. ■filecite■turn7file20■L12-L33■ ■filecite■turn7file20■L118-L156■

The organizer guide says the selected problem determines the algorithm, instances, metrics, and starter materials. It explicitly lists S3 as “Noise in Trotter Simulation.” ■filecite■turn7file20■L36-L40■ ■filecite■turn7file20■L62-L90■

The hackathon overview slide describes Processor A as a 5-qubit fixed baseline graph with synthetic noise, and Processor B as a 7-qubit heavy-hex-inspired graph with a contemporary compilation baseline. Treat that as high-level context only; Antigravity must still use the exact definitions/files in the Starter Guide, not recreate them from the picture. ■filecite■turn7file19■L49-L67■

2. Before Antigravity edits anything

1. Open Antigravity and open the existing FloquetSim Explorer project folder.
2. Create a new branch or backup before making major edits.
3. Confirm the official selected problem code. If S3 is not your registered problem, stop and adapt the project to the correct one.
4. Download the official S3 Starter Guide and all accompanying processor/configuration files from the organizer folder.
5. Put those files in a clearly named folder in the project, such as official_materials/ (unless the organizers specify another structure).
6. Tell Antigravity not to overwrite or delete the original project and not to make assumptions where official files are missing.

> Processor A and B are organizer-defined benchmark models/configurations; do not assume that they are physical quantum computers you must personally access. The guide determines whether to simulate them, transpile against them, or use another specified workflow. Do not hardcode illustrative topologies as if they were official.

3. Main Antigravity prompt — paste this into the project

Copy the prompt below into Antigravity after opening the project and adding the official Starter Guide/configuration files. If you have not obtained the official processor definitions, do not ask Antigravity to guess them; ask it to identify exactly which files are missing.

```
You are the senior quantum software engineer assisting a beginner participant in Qiskit Fall Fest 2026 Phase 1.

GOAL
Prepare a reproducible, scientifically honest Phase 1 submission in a NEW GitHub repository, based on my existing Floque

NON-NEGOTIABLE RULES
1. Read the complete official Phase 1 instructions and the selected problem's Starter Guide before implementing.
2. Use the exact organizer-supplied Processor A and Processor B definitions and the exact problem-specific algorithm, pa
3. Never invent a processor configuration, coupling map, noise model, benchmark value, measurement, citation, deadline,
4. Never label simulated results as hardware results. Clearly state whether results are simulated, analytically computed
5. Do not claim a run passed unless you ran it and inspected its outputs.
6. Keep a backup/branch and do not destroy unrelated existing work.
7. Do not add a website feature unless it directly helps the required scientific deliverable. Prioritize correct experim
8. If an essential official file or instruction is missing/ambiguous, stop at that dependency, explain the exact missing
9. Work in small phases. At the end of each phase, report files changed, commands run, results, and blockers. Do not sil

PHASE A — AUDIT
- Inspect the repository, current branch, README, dependency files, source, notebooks, tests, and existing results.
- Do not modify files yet.
- Summarize what is implemented, what is untested, what is missing, and whether the current code matches the official S3
- Identify the official Processor A/B definition files and quote their actual paths. If not present, stop and request th
- Propose a safe plan for a separate submission repository.

PHASE B — REQUIREMENTS MAP
Create a requirements checklist mapping each official requirement to: source document/section, implementation file, outp
Extract the exact S3 algorithm and controlled experiment from the official guide. Do not substitute a generic Trotter si
Extract Processor A/B graph, qubit count, gate basis, noise assumptions, transpilation settings, seeds/shots, and requir

PHASE C — IMPLEMENTATION
- Implement the official S3 workflow, reusing valid code from FloquetSim Explorer where appropriate.
- Use the official A and B definitions without modifying them unless the guide explicitly allows a change.
- Make the A/B runs use the same controlled problem instance, algorithm settings, and fair comparison conditions.
- Keep ideal/reference and noisy cases separate if required by the guide.
- Add deterministic seeds and configuration logging wherever permitted and appropriate.
- Save raw results and processed summaries in machine-readable files.
- Make sure output values are computed from the run, not hardcoded or copied from examples.
```

PHASE D — METRICS AND ANALYSIS
Follow the official guide's metric definitions. The Phase 1 overview identifies solution/problem-specific output, circui
Produce A-vs-B tables and clear graphs from actual output files. Explain the observed differences and limitations withou

PHASE E — REPRODUCIBILITY
- Provide a clean setup path, supported Python version, dependency/environment file, and exact run commands.
- Add checks that fail loudly if official processor definitions or required outputs are absent.
- Run the documented experiment for A and B and any available tests.
- Check that expected output files were newly generated and contain valid data.
- Inspect at least one row/value from each results table and verify every generated figure exists and corresponds to the
- If execution fails, report the exact error and fix it if possible. Never fabricate successful results.

PHASE F — SUBMISSION REPOSITORY
Create a NEW repository directory for submission, separate from the current website/project if practical. Include only t
- main.py or main.ipynb with end-to-end implementation
- official Processor A/B files or references/copies as permitted by competition rules
- results/ containing actual tables, graphs, and exported data
- README.md with problem code/title, setup, exact run commands, methodology, metrics, A/B findings, limitations, and rep
- requirements.txt, environment.yml, pyproject.toml, or the organizer-required environment specification
- report.pdf or a complete analysis section in the notebook
- AI-use disclosure if required by the organizers
Follow any official repository template/naming convention if supplied; it overrides this suggested structure.

PHASE G — FINAL VERIFICATION AND GIT
- Show git status and current branch before committing.
- Never include secrets, credentials, private data, cache files, or irrelevant website build output.
- Run the full documented workflow from a clean environment if feasible.
- Check that the README commands match actual commands and paths.
- Prepare a final checklist with PASS / FAIL / BLOCKED for every official requirement.
- Commit and push only after showing me the planned files and summarizing verification results. Do not claim the reposit
- Do not fill out or submit an external competition form unless I explicitly authorize that action and provide any requi

FINAL RESPONSE FORMAT
1. What is complete
2. Exact commands run and whether they passed
3. Actual Processor A results and source files
4. Actual Processor B results and source files
5. A/B comparison with generated tables/figures
6. Files created/changed
7. Remaining blockers or uncertainties
8. GitHub repository/branch status
9. Final submission checklist
10. Any task that requires my action

Start with PHASE A only. Audit first; do not edit code until you have reported the audit and confirmed the official prob

4. What the processor definitions should tell you

The official S3 guide may provide JSON/YAML/Python definitions, a notebook, or written instructions. Antigravity must locate the real files and identify these properties where supplied:

Item	What to verify
Processor A	Exact graph/coupling map, number of qubits, supported basis gates, noise assumptions, and any compilation settings.
Processor B	The same categories, using the official B definition.
Problem instance	Exact Hamiltonian/system, parameters, initial state, evolution time, Trotter steps/order, and other S3 requirements.
Execution model	Whether the guide requires local simulation, noise simulation, cloud/QPU execution, or another method.
Fair comparison	Which settings must remain identical and which processor-specific settings may differ.
Metrics	Exact definition and units for problem quality, depth, two-qubit gates, SWAPs/routing, qubits, and noise/reproducibility.

> Do not create processor_A.json or processor_B.json just because those filenames appear in an example. Actual filenames and formats come from the organizer bundle.

5. Expected submission structure

Use the official repository template if the organizers supplied one. Otherwise, this is a sensible starting structure; Antigravity should adapt it to the real project and rules:

```
qff-phase1-s3-submission/
■■■ README.md
■■■ main.py                 # or main.ipynb
■■■ requirements.txt        # or official environment file
■■■ processors/             # exact official A/B files, if allowed/required
■■■ src/                    # reusable implementation modules, if needed
■■■ results/
■   ■■■ raw/
■   ■■■ tables/
■   ■■■ figures/
■■■ report.pdf              # or analysis in notebook
■■■ AI_DISCLOSURE.md        # only if required by policy
```

The Phase 1 instructions list an end-to-end script/notebook, Processor A/B definitions or required files, results, README, report/analysis, environment specification, and figures. ■filecite■turn7file20■L144-L171■

6. Required result comparison

At minimum, prepare a table with one row per processor and the problem-specific metric(s) required by S3. Also include the common architecture/resource measures required by the official guide. The organizer's instructions identify solution quality, circuit depth, two-qubit gate count, SWAP/routing overhead, physical qubit use, and noise/reproducibility as comparison areas. ■filecite■turn7file20■L118-L135■

Measurement	Processor A	Processor B	How obtained
S3-specific output	Actual run	Actual run	As defined by S3 guide
Circuit depth	Actual transpiled circuit	Actual transpiled circuit	Same counting convention
Two-qubit gate count	Actual count	Actual count	After required compilation stage
SWAP/routing overhead	Actual count/metric	Actual count/metric	Report if required/applicable
Physical qubits	Actual use	Actual use	Include ancillas if applicable

Measurement	Processor A	Processor B	How obtained
Noise effect / repeatability	Measured result	Measured result	State shots, seeds, noise assumptions

7. GitHub: create a separate submission repository

The user requested a new repository for the final submission. Antigravity should create a clean submission folder based on the existing project, without changing the existing repository's identity or deleting the original. First verify the official naming convention. If no naming convention is supplied, propose a clear name and ask before creating a remote repository if the tool requires account permission.

Example terminal sequence, only after the files have been reviewed and tests/results have been verified:

```
git status
git branch --show-current
# Review the exact files that will be included before adding them.
git add README.md main.py requirements.txt processors/ results/ report.pdf
git status
git commit -m "Add Phase 1 Processor A-B benchmark"
# Add/configure the NEW remote only when its URL is known:
git remote -v
git push -u origin main
```

These are examples, not commands to run blindly. The actual main file, environment file, processor path, branch name, remote, and repository naming convention may differ. Never push credentials or private tokens. Verify the repository page after the push.

8. Submission form and final checks

The supplied Phase 1 instructions say the submission form may ask for participant/team identity, problem code, GitHub repository URL, a Hugging Face Space URL if applicable, and confirmation that judges can access the repository and that the correct version is submitted. The official submission form and current deadline must be taken from organizer announcements, not guessed. ■filecite■turn7file20■L174-L182■

Before submission, confirm:

■ Exactly one official problem code is used.
■ Official S3 guide and A/B definitions are in hand.
■ Processor A experiment completed and output inspected.
■ Processor B experiment completed and output inspected.
■ Results are genuine and reproducible; no invented values.
■ Required metrics and A/B comparison are present.
■ Graphs and tables are generated from result files.
■ README and environment setup are accurate.
■ Notebook/script runs from a clean setup where feasible.
■ New repository is accessible to judges.
■ AI disclosure included if required.
■ Correct repository URL/version submitted using the official form.

9. What Antigravity must not do

• Do not fabricate Processor A/B definitions or use the diagrams in an overview slide as a substitute for official machine-readable files.
• Do not fabricate results, figures, metrics, or successful test logs.
• Do not silently switch from S3 to another problem statement.
• Do not claim a GitHub push equals competition submission.
• Do not submit without checking that the repository is accessible and the required files exist.
• Do not remove the existing website/project unless explicitly asked.
• Do not continue past a missing official configuration; report the blocker and request the exact file.

10. The exact next action for you

Step 1: Open Antigravity and load your existing FloquetSim Explorer project.
Step 2: Open the official Starter Guide folder: https://drive.google.com/drive/folders/1caxlXXeqjOkLmXGlimy_Etw0uPHSwyga
Step 3: Download the S3 guide and all official Processor A/B definitions or starter files.
Step 4: Place the files in the project, or upload them into Antigravity's workspace if the tool supports that.
Step 5: Paste the main Antigravity prompt from Section 3.
Step 6: Let it audit first. If it says a processor file is missing, do not tell it to guess; provide the missing file.

Step 7: Let it implement and run the verified experiments, then review the PASS/FAIL/BLOCKED checklist before submitting.

Final reminder

This guide is designed to make Antigravity do the development work in a structured way, but no tool can responsibly guarantee a valid competition submission without checking actual execution and required files. The official instructions emphasize reproducibility, correct results, architecture comparison, and clear documentation.
■filecite■turn7file20■L184-L213■